Lab-sheet 4:

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Design and execute SQL commands to **create users, manage their access privileges, and remove users** in a database system. This assignment focuses on **database security and access control mechanisms**.

Part A: User Creation

1. Create a new user named user1 with a password.
2. Create another user user2 with limited access.

Part B: Granting Privileges

3. Grant **SELECT and INSERT** privileges on a table Student to user1.
4. Grant **ALL privileges** on a table Course to user2.
5. Provide **WITH GRANT OPTION** to user1 and observe its effect.

Part C: Revoking Privileges

6. Revoke **INSERT permission** from user1 on the Student table.
7. Revoke all privileges from user2 on the Course table.

Part D: User Deletion

8. Drop/Delete user user2 from the database.
9. Attempt to login using deleted credentials and observe the result.

Part E: Additional Commands

10. Display all users in the system.
11. Show privileges assigned to a specific user.

SQL QUERIES:

```sql
CREATE DATABASE labsheet4;
USE  labsheet4;

CREATE TABLE Student (
    id INT PRIMARY KEY,
    name VARCHAR(50),
    age INT
);
INSERT INTO Student VALUES
(1, 'Amit', 20),
(2, 'Neha', 21);

CREATE TABLE Course (
    course_id INT PRIMARY KEY,
    course_name VARCHAR(50)
);

INSERT INTO Course VALUES
(101, 'DBMS'),
(102, 'OS');
```

Part A:

```sql
CREATE USER 'user1'@'localhost' IDENTIFIED BY 'pass123';
CREATE USER 'user2'@'localhost' IDENTIFIED BY 'pass123';
```

Part B:

```sql
GRANT SELECT, INSERT ON labsheet4.Student TO 'user1'@'localhost';
GRANT ALL PRIVILEGES ON labsheet4.Course TO 'user2'@'localhost';

GRANT SELECT, INSERT ON labsheet4.Student
TO 'user1'@'localhost' WITH GRANT OPTION;
```

Part C:

```sql
SHOW GRANTS FOR 'user1'@'localhost';
REVOKE INSERT ON labsheet4.Student FROM 'user1'@'localhost';

REVOKE ALL PRIVILEGES ON labsheet4.Course FROM 'user2'@'localhost';
SHOW GRANTS FOR 'user1'@'localhost';
```

Output:

```
30
31 •  SHOW GRANTS FOR 'user1'@'localhost';
32 •  REVOKE INSERT ON labsheet4.Student FROM 'user1'@'localhost';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Grants for user1@localhost
GRANT USAGE ON *.* TO `user1`@`localhost`
GRANT SELECT, INSERT ON `labsheet4`.`stud...

```
33 ⊠  REVOKE ALL PRIVILEGES ON labsheet4.Course FROM 'user2'
34 •  SHOW GRANTS FOR 'user1'@'localhost';
35
36 •  DROP USER 'user2'@'localhost';
```

esult Grid | Filter Rows: | Export: | Wrap Cell Content:

Grants for user1@localhost
GRANT USAGE ON *.* TO `user1`@`localhost`
GRANT SELECT ON `labsheet4`.`student` TO ...

Part D:

```
DROP USER 'user2'@'localhost';
SELECT user, host FROM mysql.user;
```

Part E:

```
•  SELECT user, host FROM mysql.user;
•  SHOW GRANTS FOR 'user1'@'localhost';
•  SELECT username FROM all_users;
```

Result Grid | Filter Rows:

user	host
user1	%
dept_user	localhost
mysql.infoschema	localhost
mysql.session	localhost
mysql.sys	localhost
root	localhost
user1	localhost

EXPLANATION:

- A database user is an account that can connect to the database and perform operations based on assigned privileges.
- **Privileges:**
  Privileges define what actions a user can perform on database objects such as tables. Common privileges include:
  - SELECT (read data)
  - INSERT (add data)
  - UPDATE (modify data)
  - DELETE (remove data)
- **CREATE USER:**
  Used to create a new database user with login credentials.
- **GRANT:**
  Used to assign privileges to users on specific database objects.
  Example: Giving SELECT permission on a table.
- **WITH GRANT OPTION:**
  Allows a user to further grant the same privileges to other users.
- **REVOKE:**
  Used to remove previously granted privileges from a user.
- **DROP USER:**
  Deletes a user account from the database system.

User **user1** was able to access and insert data into the Student table initially.

After revoking INSERT privilege, user1 could no longer insert data, confirming that REVOKE works correctly.

The **WITH GRANT OPTION** allowed user1 to grant permissions to other users, demonstrating privilege delegation.

User **user2** had full access to the Course table initially, but after revoking privileges, access was completely removed.

After executing **DROP USER**, user2 was no longer able to log in, confirming successful deletion.

System queries successfully displayed existing users and their privileges.